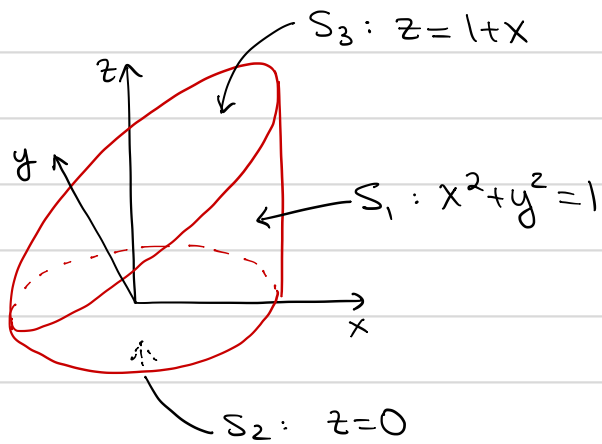
Lecture 22

Surface integrals & flux (16.7)

Example: $\iint_S z\,dS = \frac{3\pi}{2} + 0 + \sqrt{2}\pi$



$S = S_1 \cup S_2 \cup S_3$

S_1: $\vec{r}(\theta, z) = (\cos\theta, \sin\theta, z)$

$(0 \le z \le 1+\cos\theta,$

$0 \le \theta \le 2\pi)$

$$\vec{r}_\theta \times \vec{r}_z = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= (\cos\theta, \sin\theta, 0)$$

$$\iint_{S_1} z\,dS = \int_0^{2\pi}\int_0^{1+\cos\theta} z \cdot \overbrace{|\vec{r}_\theta \times \vec{r}_z|}^{1}\,dz\,d\theta$$

$$= \int_0^{2\pi} \frac{(1+\cos\theta)^2}{2}\,d\theta$$

$$= \cdots = \frac{3\pi}{2} \;//$$

S_2: $\iint_{S_2} z\,dS = \iint_{S_2} 0\,dS = 0$

S_3: the graph of $f(x,y) = 1+x$ over $D = \{x^2+y^2 \le 1\}$

$$\iint_{S_3} z\,dS = \iint_D z\sqrt{f_x^2 + f_y^2 + 1}\,dx\,dy$$

$$= \iint_D (1+x)\sqrt{1+0+1}\,dx\,dy$$

$$= \int_0^{2\pi}\int_0^1 (1+\cos\theta)\sqrt{2}\,r\,dr\,d\theta$$

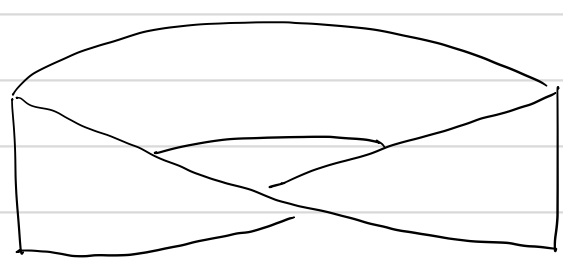
$$= \sqrt{2}\cdot \frac{r^2}{2}\Big|_0^1 \cdot \theta + \sin\theta\Big|_0^{2\pi}$$

$$= \sqrt{2}\cdot\pi$$

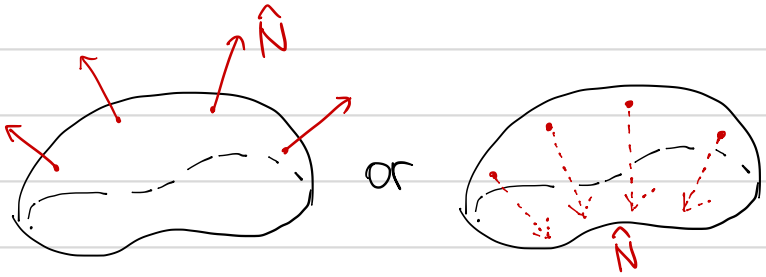
Orientations

A surface is <u>orientable</u> if it has two sides.

Example: The Möbius strip is not orientable.



An <u>orientation</u> of a surface S is a continuous choice of normal vectors on S.



$\vec{r}(u,v) = \langle x(u,v), y(u,v), z(u,v) \rangle$

$$\hat{N} = \pm \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} \quad \text{unit normal}$$

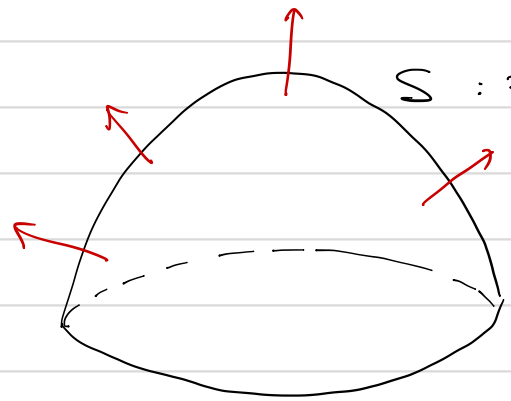
<u>Integral of a vector field $\vec{F}$ on an oriented surface S</u>

$$\boxed{\iint_S \vec{F} \cdot d\vec{S} := \iint_S \vec{F} \cdot \hat{N} \, dS}$$

"Flux of $\vec{F}$ across S"

$$d\vec{S} = \hat{N} \, dS$$
$$= \pm \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} |\vec{r}_u \times \vec{r}_v| \, du \, dv$$
$$= \pm (\vec{r}_u \times \vec{r}_v) \, du \, dv$$

<u>Example</u>: $\vec{F} = (P, Q, R) = (y, x, z)$



$S : z = \underbrace{1 - x^2 - y^2}_{f(x,y)}$

On S: $\vec{r}(x,y) = (x, y, f(x,y))$

$$\iint_{S_1} \vec{F} \cdot \underbrace{\vec{N} \, dS}_{d\vec{S}} = \iint_D \vec{F} \cdot (\vec{r}_x \times \vec{r}_y) \, dx \, dy$$

$$\begin{aligned}
\vec{F} \cdot (\vec{r}_x \times \vec{r}_y) &= (P, Q, R) \cdot (-f_x, -f_y, 1) \\
&= -P f_x - Q f_y + R \\
&= -y \cdot (-2x) - x(-2y) + z \\
&= 4xy + 1 - x^2 - y^2
\end{aligned}$$

$$\iint_{S_1} \vec{F} \cdot d\vec{S} = \iint_D (4xy + 1 - x^2 - y^2) \, dA$$

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^{1} (4r^2 \cos\theta \sin\theta + 1 - r^2) \cdot r \, dr \, d\theta$$

$$= \cdots = \frac{\pi}{2} \; //$$